Edward Zhu

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Education

University of California, Irvine Sep. 2023 – Dec. 2026
B.S. in Computer Science, Minor in Mathematics - GPA: 3.94 *Irvine, CA*

Experience

SWE Intern - Dropbox June 2026 – Sep. 2026
Core-Sync team
- Developed a **full-stack** system allowing team admins to apply setting updates to existing users devices
- Designed and implemented an idempotent, backward-compatible delivery model and database schema
- Produced safe production code utilizing **GrowthBook** for controlled feature testing and gradual rollout
- Pushed end-to-end changes across **React, Python, Go, Rust, and Protobuf APIs**

SWE Intern - GreenPeak.AI June 2025 – Sep. 2025
Startup working on automatic AI content generation
- Engineered a **Model Context Protocol (MCP) Server** streamlining LLM integration with internal tools
- Built an automated media generation pipeline for producing music and video assets and uploading to Youtube
- Automated web interactions and content extraction using **Playwright**
- Created a **Google OAuth** token management system

Competitive & Technical Achievements

Collegiate Penetration Testing Competition (CPTC) - Global Champion 2025-Present
- Placed **first** at CPTC Global Finals on a team of six
- Orchestrated a **multi agent** penetration testing pipeline using **Claude Code**
- Developed **bash** scripts to automate enumeration, tooling setup, and exploitation
- Delivered a **120** page technical report and gave executive presentation on discovered findings

Collegiate Cyber Defense Competition (CCDC) National Champions - Team Alternate 2024-2025
- Sponsored for victors tour of Pentagon, CIA and FBI headquarters and met with NSA and CISA executives
- Defended against red team attacks proactively by implementing **host-based firewalls** and patches
- Deployed **dockerized** services and modified them through YAML-based configurations and volumes

Projects

Typst Collaborative Document Editor | *Node.js, Rust, Yjs, WebSockets, Docker* 2026
Built a self-hosted real-time collaborative **Typst** editor with a live multi-user preview
- Implemented conflict-free multi-user editing with **Yjs CRDTs** over **WebSockets**
- Engineered an incremental compile engine in **Rust** that content-hashes pages and renders changed ones in **parallel**
- Designed **git**-backed version history, exposing every project as a read-only clonable repository over **HTTP**
- Automated cross-platform setup on **macOS** and **Debian** with a provisioning script and a **Docker** deployment

RAG Google Drive Document Assistant | *Python, Pandas, Google API, OpenAI API* 2024
Developed an end-to-end **RAG** system that extracts insights from a user's relevant Google Docs
- Leveraged the **Google Drive API** to store documents in **CSV** files
- Generated **vector embeddings** of document content using OpenAI's Embedding API for semantic understanding
- Implemented a vector search pipeline to identify and rank relevant documents based on user queries
- Integrated **OpenAI's GPT API** to generate context-aware responses grounded in retrieved document content

Certificates and Awards

- MIT Data Science and ML
- APIsec Certified Practitioner (ACP)

Technical Skills

Languages: Python, C/C++, SQL, Java, Bash, PowerShell, MIPS Assembly, Lisp, Prolog
Developer Tools: Git, GDB, Valgrind, Docker, VS Code, PyCharm, Vim
Other: Pandas, Keras, OpenCV2, Windows, Linux, Active Directory, Metasploit, NetExec, Burp Suite